

Machine Learning for teens

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table of contents

- ▶ **Gradient Decent – Cost Function**
- ▶ **Handling categorical Features**
- ▶ **Managing missing data**

Gradient Decent - Cost Function

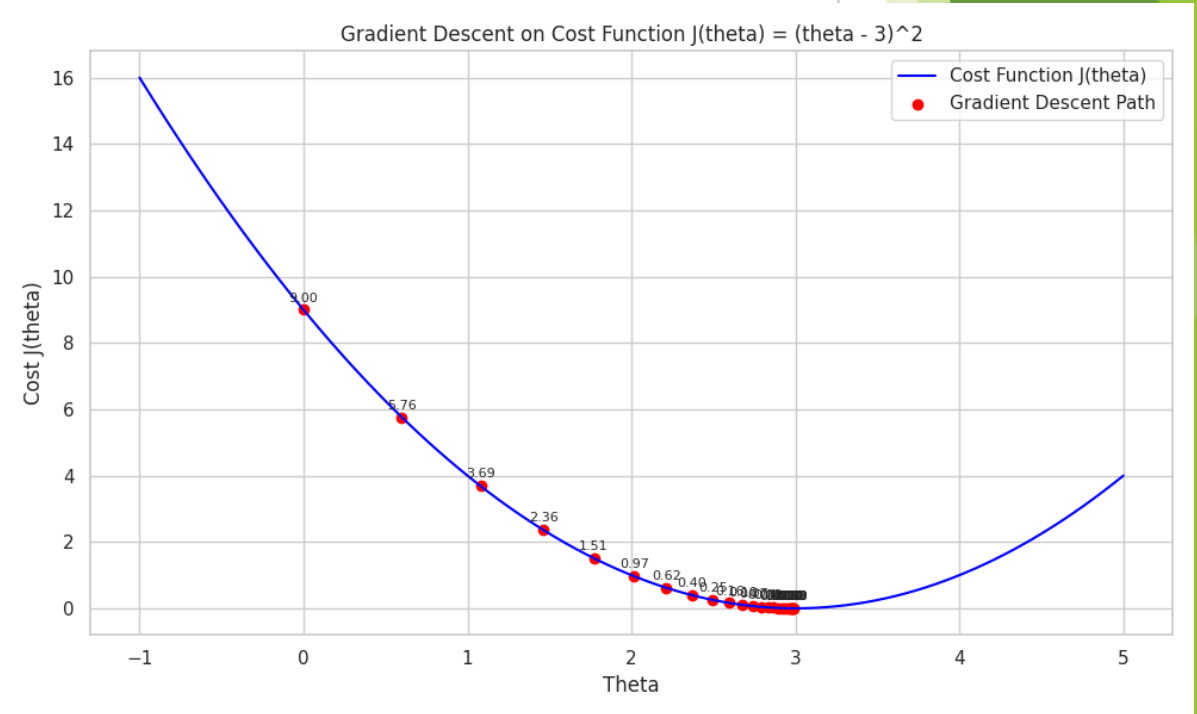
- ▶ The goal of machine learning is to reduce difference between our predicted values and our actual values(or minimizing loss function)
- ▶ What is cost function or loss function
 - ▶ Difference between actual data and model's guess
 - ▶ There more loss functions but some of them are like
 - ▶ In regression task(continuous target variable)
 - ▶ MSE (Mean Squared Error)
 - ▶ MAE (Mean Absolute Error)
 - ▶ In classification task (discrete target variable)
 - ▶ Cross-Entropy Loss

Gradient decent

- This is a method to minimized cost function

$$\theta = \theta - \alpha \cdot \frac{\partial J(\theta)}{\partial \theta}$$

- θ : model parameters
- α : learning rate
- $J(\theta)$: cost function
- $\frac{\partial J(\theta)}{\partial \theta}$: gradient (slope)



Example for gradient decent

- ▶ **Step 1: Setup**

- ▶ $\hat{y}=mx+b$

x	y
1	2
2	4

- ▶ **Step 2: Initialize**

- ▶ Start with:

- ▶ $m=0$

- ▶ $b=0$

- ▶ Learning rate $\alpha=0.1$

Example continued

► Step 3: Compute Predictions

- Using $m=0$, $b=0$:
 - $\hat{y}^1=0$ $\hat{y}^2=0$

$$\text{MSE} = \frac{1}{2}[(y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2]$$

► Step 4: Compute Loss (MSE)

- $\text{MSE} = 1/2[(2-0)^2 + (4-0)^2] = 1/2(4+16) = 10$

► Step 5: Compute Gradients

- We'll use the partial derivatives of the cost function:

$$\frac{\partial J}{\partial m} = \frac{-2}{n} \sum x_i(y_i - \hat{y}_i) = \frac{-2}{2}[1(2 - 0) + 2(4 - 0)] = -2(2 + 8) = -20$$

$$\frac{\partial J}{\partial b} = \frac{-2}{n} \sum (y_i - \hat{y}_i) = \frac{-2}{2}[2 - 0 + 4 - 0] = -2(6) = -12$$

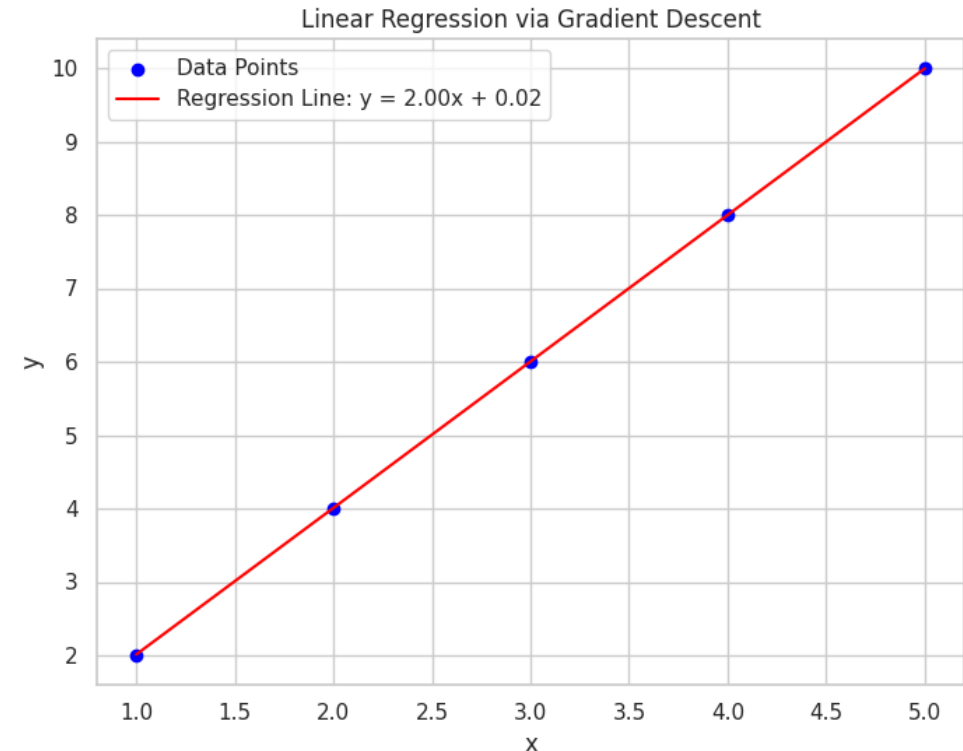
Example continued

► Step 6: Update Parameters

$$m = m - \alpha \cdot \frac{\partial J}{\partial m} = 0 - 0.1 \cdot (-20) = 2 \quad b = b - \alpha \cdot \frac{\partial J}{\partial b} = 0 - 0.1 \cdot (-12) = 1.2$$

► Step 7: Repeat

- Now use $m=2, b=1.2$ to compute new predictions, loss, gradients, and update again. Each epoch (full pass through the data) brings you closer to the best line.
- For more detail see code [Gradient Decent](#)



Handling Categorical Features

- ▶ Basically, we have two value types for a column
 - ▶ Categorical
 - ▶ Good, bad – hot ,cold
 - ▶ Numerical
 - ▶ 1,4,2,5,6
 - ▶ Boolean (can treat as numerical)
 - ▶ True , False
 - ▶ Datetime
 - ▶ 2025-09-01 14:30:00
 - ▶ Text (object)
 - ▶ Comment, review
 - ▶ Excellent analysis and clear writing.

Handling Categorical Features

- ▶ Computer can't understand categorical data, we need to convert them into numerical
- ▶ Two types of categorical data
 - ▶ Ordinal
 - ▶ There is order between data
 - ▶ Low , medium and high
 - ▶ Nominal
 - ▶ Just labels , no order
 - ▶ Red, blue and green

Handling Categorical Features

► Categorical Features types

► ordinal

► label encoding or ordinal encoding

- Avoid one-hot encoding because remove order info
- Use ordinal encoding because it saved relationship between columns

► nominal

► one-hot encoding or dummy encoding

- Don't use label encoding or ordinal because model consider order that doesn't exist

Managing missing data

- ▶ What missing data means? Every cell without value
- ▶ How to handle it?
 - ▶ ◆ 1. Drop Missing Data
 - ▶ Only a small portion of data is missing and it's not critical
 - ▶ ◆ 2. 1. Fill with Mean, Median, or Mode
 - ▶ Use when: Data is numerical and missing at random
 - ▶ ◆ 2.2. and for categorical data replace with a string like letter o
- ▶ Load dataset(student_grades_modified.csv)
- ▶ Find missing value
- ▶ Handle missing data

Assignment

- ▶ Handel missing data and categorical data for [student_grades_modified.csv](#)